

Central (spherically symmetric) potential

$$\underline{3D} \quad \hat{p} = \frac{\hbar}{i} \vec{\nabla} = \frac{\hbar}{i} \left(\frac{\partial}{\partial x_1}, \frac{\partial}{\partial x_2}, \frac{\partial}{\partial x_3} \right) = (\hat{p}_1, \hat{p}_2, \hat{p}_3)$$

$$\hat{x} = (\hat{x}_1, \hat{x}_2, \hat{x}_3)$$

$$[\hat{x}_i, \hat{p}_i] = i\hbar, \quad i = 1, 2, 3$$

$$\underline{\text{Sch eqn}}: -\frac{\hbar^2}{2m} \nabla^2 \psi(\vec{x}) + V(\vec{x}) \psi(\vec{x}) = E \psi(\vec{x})$$

$V = V(|\vec{x}|) = V(r) \rightarrow$ spherically symmetric
invariant under rotations
 $\rightarrow$ central potential

$$\nabla^2 \psi = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \psi}{\partial r} \right) + \frac{1}{r^2} \left(\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \sin \theta \frac{\partial}{\partial \theta} + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \varphi^2} \right) \psi$$

$$\psi = \psi(\vec{x})$$

$V = V(r)$, Also when the potential between two particles depend only on the distance between the particles

$V(\vec{x}_1, \vec{x}_2)$ is a function of $|\vec{x}_1 - \vec{x}_2|$ only.

Sch. eqn. for a particle in a central potential

$$-\frac{\hbar^2}{2m} \left[\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \psi}{\partial r} \right) + \frac{1}{r^2} \left(\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \sin \theta \frac{\partial}{\partial \theta} + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \varphi^2} \right) \right] \psi + V(r) \psi = E \psi$$

where, $\psi = \psi(r, \theta, \varphi)$

Particle rotates around a fixed point

→ orbital angular momentum $\vec{L} = \vec{r} \times \vec{p}$

$$\left. \begin{aligned} \hat{L}_x &= \hat{y} \hat{p}_z - \hat{z} \hat{p}_y \\ \hat{L}_y &= \hat{z} \hat{p}_x - \hat{x} \hat{p}_z \\ \hat{L}_z &= \hat{x} \hat{p}_y - \hat{y} \hat{p}_x \end{aligned} \right\} \hat{L}_i \rightarrow \text{Hermitian operators} \\ \rightarrow \text{observable.}$$

Also $[x_i, p_j] = i\hbar \delta_{ij}$
If $i \neq j$, ordering does not matter.

$$[\hat{L}_x, \hat{L}_y] = [\hat{y} \hat{p}_z - \hat{z} \hat{p}_y, \hat{z} \hat{p}_x - \hat{x} \hat{p}_z]$$

[Use the identity $[AB, C] = A[B, C] + [A, C]B$,

$$[A, BC] = B[A, C] + [A, B]C]$$

$$\begin{aligned} \Rightarrow [\hat{L}_x, \hat{L}_y] &= [\hat{y} \hat{p}_z, \hat{z} \hat{p}_x] + [\hat{z} \hat{p}_y, \hat{x} \hat{p}_z] \\ &= [\hat{y} \hat{p}_z, \hat{z}] \hat{p}_x + \hat{x} [\hat{z} \hat{p}_y, \hat{p}_z] \\ &= \hat{y} [\hat{p}_z, \hat{z}] \hat{p}_x + \hat{x} [\hat{z}, \hat{p}_z] \hat{p}_y \\ &= \hat{y} (-i\hbar) \hat{p}_x + \hat{x} (i\hbar) \hat{p}_y \\ &= i\hbar (\hat{x} \hat{p}_y - \hat{y} \hat{p}_x) = i\hbar \hat{L}_z \end{aligned}$$

$$\Rightarrow \boxed{[\hat{L}_x, \hat{L}_y] = i\hbar \hat{L}_z}$$

similarly, $\boxed{[\hat{L}_y, \hat{L}_z] = i\hbar \hat{L}_x}$

$$\boxed{[\hat{L}_z, \hat{L}_x] = i\hbar \hat{L}_y}$$

Commutators of angular momenta give another angular momenta!

$\hat{L}_x, \hat{L}_y, \hat{L}_z$ cannot have simultaneous eigenstates

Define $\hat{L}^2 = \hat{L}_x \hat{L}_x + \hat{L}_y \hat{L}_y + \hat{L}_z \hat{L}_z$

$$[\hat{L}_z, \hat{L}^2] = [\hat{L}_z, \hat{L}_x \hat{L}_x] + [\hat{L}_z, \hat{L}_y \hat{L}_y]$$

$$= [\hat{L}_z, \hat{L}_x] \hat{L}_x + \hat{L}_x [\hat{L}_z, \hat{L}_x]$$

$$+ [\hat{L}_z, \hat{L}_y] \hat{L}_y + \hat{L}_y [\hat{L}_z, \hat{L}_y]$$

$$= i\hbar \hat{L}_y \hat{L}_x + i\hbar \hat{L}_x \hat{L}_y + (-i\hbar \hat{L}_x) \hat{L}_y + \hat{L}_y (-i\hbar \hat{L}_x)$$

$$= 0$$

$\hat{L}_z, \hat{L}^2 \rightarrow$ have simultaneous eigenstates

Similarly, $[\hat{L}_x, \hat{L}^2] = [\hat{L}_y, \hat{L}^2] = 0$

Position space representation of $\hat{L}_i$ operators

$$\left. \begin{aligned} x &= r \sin \theta \cos \phi \\ y &= r \sin \theta \sin \phi \\ z &= r \cos \theta \end{aligned} \right\} \begin{aligned} r &= \sqrt{x^2 + y^2 + z^2} \\ \theta &= \cos^{-1} \frac{z}{r} \\ \phi &= \tan^{-1} \frac{y}{x} \end{aligned}$$

$$\begin{aligned} \frac{\partial}{\partial \phi} &= \frac{\partial x}{\partial \phi} \frac{\partial}{\partial x} + \frac{\partial y}{\partial \phi} \frac{\partial}{\partial y} + \frac{\partial z}{\partial \phi} \frac{\partial}{\partial z} \\ &= (-y) \frac{\partial}{\partial x} + x \frac{\partial}{\partial y} \\ &= x \frac{\partial}{\partial y} - y \frac{\partial}{\partial x} \end{aligned}$$

Again, $\hat{L}_z = \frac{\hbar}{i} \left(x \frac{\partial}{\partial y} - y \frac{\partial}{\partial x} \right) = \frac{\hbar}{i} \frac{\partial}{\partial \phi}$

Similarly, $\hat{L}_x = -\frac{\hbar}{i} \left(\sin \theta \frac{\partial}{\partial \theta} + \cot \theta \cos \phi \frac{\partial}{\partial \phi} \right)$

$$\hat{L}_y = \frac{\hbar}{i} \left(\cos \theta \frac{\partial}{\partial \theta} - \cot \theta \sin \phi \frac{\partial}{\partial \phi} \right)$$

$\hat{L}_i$ involves derivatives w.r.t angular co-ordinates only

$$\begin{aligned}\hat{L}^2 &= \hat{L}_x \hat{L}_x + \hat{L}_y \hat{L}_y + \hat{L}_z \hat{L}_z \\ &= -\hbar^2 \left(\frac{1}{\sin\theta} \frac{\partial}{\partial\theta} \sin\theta \frac{\partial}{\partial\theta} + \frac{1}{\sin^2\theta} \frac{\partial^2}{\partial\varphi^2} \right)\end{aligned}$$

Let $\psi_{lm}(\theta, \varphi)$ are the simultaneous eigenstates of $\hat{L}_z, \hat{L}^2$.

$$\begin{cases} \hat{L}_z \psi_{lm}(\theta, \varphi) = m\hbar \psi_{lm}(\theta, \varphi), \quad m \text{ is real} \\ \hat{L}^2 \psi_{lm}(\theta, \varphi) = l(l+1)\hbar^2 \psi_{lm}(\theta, \varphi), \quad l \text{ is real} \end{cases}$$

$$\frac{\hbar}{i} \frac{\partial}{\partial\varphi} \psi_{lm}(\theta, \varphi) = m\hbar \psi_{lm}(\theta, \varphi)$$

$$\Rightarrow \frac{\partial \psi_{lm}}{\partial\varphi} = im \psi_{lm} \Rightarrow \psi_{lm}(\theta, \varphi) = e^{im\varphi} P_l^m(\theta)$$

$$\text{Demand } \psi_{lm}(\theta, \varphi + 2\pi) = \psi_{lm}(\theta, \varphi)$$

$$\Rightarrow e^{im(\varphi + 2\pi)} = e^{im\varphi} \Rightarrow e^{2\pi im} = 1$$

$$m = 0, \pm 1, \dots$$

[We can also show that eigenvalues of $\hat{L}_z$ are $m\hbar$
Let the φ dependent eigenfunction be $f(\varphi)$

$$\hat{L}_z f(\varphi) = \lambda f(\varphi) \quad \lambda \text{ is the eigenvalue to be determined}$$

$$\Rightarrow \frac{df}{d\varphi} - \frac{i\lambda}{\hbar} f = 0$$

$$\Rightarrow f(\varphi) = c e^{(i\lambda/\hbar)\varphi}$$

$$f(\varphi + 2\pi) = f(\varphi) \Rightarrow e^{i\frac{\lambda\varphi}{\hbar}} e^{i\frac{2\pi\lambda}{\hbar}} = e^{i\frac{\lambda\varphi}{\hbar}}$$

$$\Rightarrow e^{i2\pi\lambda/\hbar} = 1 \Rightarrow \lambda = m\hbar$$

$$m = 0, \pm 1, \pm 2, \dots]$$

Eigenvalues of $\hat{L}_z$ are $m\hbar$ ($m = 0, \pm 1, \dots$)

Eigenfunctions are $e^{im\varphi}$

Normalize the eigenfunction $\int_0^{2\pi} |f(\varphi)|^2 d\varphi = 1$

$$\text{where } f(\varphi) = c e^{im\varphi} \Rightarrow c = \frac{1}{\sqrt{2\pi}}$$

The normalized eigenfunction of $\hat{L}_z$: $\frac{1}{\sqrt{2\pi}} e^{im\varphi}$

Again, recall $\hat{L}^2 = -\hbar^2 \left(\frac{1}{\sin\theta} \frac{\partial}{\partial\theta} \sin\theta \frac{\partial}{\partial\theta} + \frac{1}{\sin^2\theta} \frac{\partial^2}{\partial\varphi^2} \right)$

$$\hat{L}^2 \psi_{lm} = l(l+1)\hbar^2 \psi_{lm}$$

$$\Rightarrow -\hbar^2 \left(\frac{1}{\sin\theta} \frac{\partial}{\partial\theta} \sin\theta \frac{\partial}{\partial\theta} + \frac{1}{\sin^2\theta} \frac{\partial^2}{\partial\varphi^2} \right) \psi_{lm} = l(l+1)\hbar^2 \psi_{lm}$$

$$\Rightarrow \left[\sin\theta \frac{\partial}{\partial\theta} \left(\sin\theta \frac{\partial}{\partial\theta} \right) + \frac{\partial^2}{\partial\varphi^2} \right] \psi_{lm} = -l(l+1) \sin^2\theta \psi_{lm}$$

$$\text{put } \psi_{lm}(\theta, \varphi) = e^{im\varphi} P_l^m(\theta)$$

$$\frac{\partial^2}{\partial\varphi^2} (e^{im\varphi}) = (im)^2 e^{im\varphi} = -m^2 e^{im\varphi}$$

$$\Rightarrow \sin\theta \frac{d}{d\theta} \left(\sin\theta \frac{dP_l^m}{d\theta} \right) - m^2 P_l^m = -l(l+1) P_l^m \sin^2\theta$$

$$\Rightarrow \sin\theta \frac{d}{d\theta} \left(\sin\theta \frac{dP_l^m}{d\theta} \right) + [l(l+1) \sin^2\theta - m^2] P_l^m = 0$$

We claim P_l^m is a function of $\cos\theta$

$$\text{put } x = \cos\theta \Rightarrow \frac{d}{d\theta} = \frac{dx}{d\theta} \frac{d}{dx} = -\sin\theta \frac{d}{dx}$$

$$\text{and } \sin\theta \frac{d}{d\theta} = -\sin^2\theta \frac{d}{dx} = -(1-x^2) \frac{d}{dx}$$

$$\Rightarrow (1-x^2) \frac{d}{dx} \left[(1-x^2) \frac{dP_l^m}{dx} \right] + [l(l+1)(1-x^2) - m^2] P_l^m(x) = 0$$

$$\Rightarrow \boxed{\frac{d}{dx} \left[(1-x^2) \frac{dP_l^m}{dx} \right] + \left[l(l+1) - \frac{m^2}{1-x^2} \right] P_l^m(x) = 0}$$

$P_l^m(x) \rightarrow$ Associated Legendre functions

For $m=0$, $P_l^0(x) = P_l(x) \Rightarrow$

$$\boxed{\frac{d}{dx} \left[(1-x^2) \frac{dP_l}{dx} \right] + l(l+1)P_l(x) = 0}$$

$P_l(x) \rightarrow$ Legendre polynomial

$$P_0(x) = 1, \quad P_1(x) = x, \quad P_2(x) = \frac{1}{2}(3x^2 - 1)$$

Let us take the soln $P_2(x)$

$$\frac{d}{dx} \left[(1-x^2) \frac{dP_2}{dx} \right] + \lambda P_2(x) = 0$$

$$\Rightarrow \frac{d}{dx} \left[(1-x^2) 3x \right] + \lambda \frac{1}{2} (3x^2 - 1) = 0$$

$$\Rightarrow (3 - 3x^2) + \frac{\lambda}{2} (3x^2 - 1) = 0$$

$$\Rightarrow \lambda = 6 \quad \Rightarrow l(l+1) = 6 \quad \Rightarrow l = 2$$

similarly $\lambda = 2$ for $P_1(x)$ on $l=1$

$\lambda = 0$ for $l=0$

possible values of $l = 0, 1, 2, 3, \dots$

values of m for a given l : $-l \leq m \leq l$

or $m = -l, -l+1, \dots, 0, \dots, (l-1), l$

In general $\boxed{\hat{L}^2 Y_{lm}(\theta, \phi) = l(l+1)\hbar^2 Y_{lm}(\theta, \phi)}$

where $Y_{lm}(\theta, \phi)$ are called spherical harmonics

$$\boxed{Y_{lm}(\theta, \phi) = P_l^m(\cos\theta) e^{im\phi}}$$

Also $\boxed{\hat{L}_z Y_{lm}(\theta, \phi) = m\hbar Y_{lm}(\theta, \phi)}$

$$\hat{L}_x = -\frac{\hbar}{i} \left(\sin \theta \frac{\partial}{\partial \theta} + \cot \theta \cos \varphi \frac{\partial}{\partial \varphi} \right)$$

$$\hat{L}_y = \frac{\hbar}{i} \left(\cos \theta \frac{\partial}{\partial \theta} - \cot \theta \sin \varphi \frac{\partial}{\partial \varphi} \right)$$

$$\hat{L}_z = \frac{\hbar}{i} \frac{\partial}{\partial \varphi}$$

$\hat{L}_x, \hat{L}^2$ operators involve derivatives w.r.t to angular co-ordinates θ, φ only $\Rightarrow [\hat{L}_x, f(r)] = 0$

$$\text{Again } \hat{H}\psi = -\frac{\hbar^2}{2m} \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \psi}{\partial r} \right) + \frac{\hat{L}^2 \psi}{2mr^2} + V(r)\psi$$

$$\Rightarrow [\hat{L}_x, \hat{H}] = 0, \text{ since } [\hat{L}_x, f(r)] = 0$$

$$\text{and } [\hat{L}_x, \hat{L}^2] = 0$$

$$\text{Similarly } [\hat{L}^2, \hat{H}] = 0$$

For spherically symmetric potentials, $\hat{L}^2, \hat{L}_z$ and $\hat{H}$ will have common eigenfunctions

$\hat{L}^2, \hat{L}_z$ are good quantum numbers

$\hat{L}^2, \hat{L}_z$ are constants of motion since they commute with the Hamiltonian.

$$\text{Again } \hat{H} = -\frac{\hbar^2}{2m} \left[\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) + \frac{1}{r^2} \left(\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \varphi^2} \right) \right] + V(r)$$

$$\text{where } \hat{L}^2 = -\hbar^2 \left(\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \varphi^2} \right)$$

$$\hat{H} = -\frac{\hbar^2}{2mr^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) + \frac{L^2}{2mr^2} + V(r)$$

Sch. eqn.

$$-\frac{\hbar^2}{2mr^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \psi}{\partial r} \right) + \frac{L^2}{2mr^2} \psi + V(r) \psi = E \psi$$

$$\psi(\vec{r}) = \psi(r, \theta, \phi) = R(r) Y_{lm}(\theta, \phi)$$

$$-\frac{\hbar^2}{2m} \frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) Y_{lm}(\theta, \phi) + \frac{R}{2mr^2} L^2 Y_{lm}(\theta, \phi)$$

Multiply both sides by $r^2/R Y$ $+ V(r) R Y_{lm}(\theta, \phi) = E R Y_{lm}(\theta, \phi)$

$$\Rightarrow -\frac{\hbar^2}{2m} \frac{1}{R} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) + \frac{1}{2m Y} L^2 Y + V(r) r^2 = E r^2$$

$$\Rightarrow \underbrace{-\frac{\hbar^2}{2m} \frac{1}{R} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) + (V(r) - E) r^2}_{\text{depends on } r \text{ only}} = \underbrace{-\frac{1}{2m Y} L^2 Y}_{\text{depends on } \theta, \phi} = \lambda$$

$$-\frac{1}{2m} L^2 Y = \lambda Y \quad \Rightarrow \quad \lambda = -\frac{l(l+1) \hbar^2}{2m}$$

$$\Rightarrow -\frac{\hbar^2}{2m} \frac{1}{R} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) + (V(r) - E) r^2 = -\frac{l(l+1) \hbar^2}{2m}$$

$$\Rightarrow -\frac{\hbar^2}{2m} \frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) + (V(r) - E) R = -\frac{l(l+1) \hbar^2}{2mr^2} R$$

$$\Rightarrow -\frac{\hbar^2}{2m} \frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) + \left[\frac{l(l+1) \hbar^2}{2mr^2} + V(r) \right] R = E R$$

Energy E depends on the angular momentum l and on $R(r)$ through $V(r)$

Angular part of ψ for spherically symmetric system is same for all $V(r) \rightarrow$ eigenfunction of L^2 ; $Y_{lm}(\theta, \phi)$

$$-\frac{\hbar^2}{2m} \frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) + \frac{l(l+1)}{2mr^2} R + V(r) R = ER$$

$$\Rightarrow -\frac{\hbar^2}{2m} \frac{d^2 R}{dr^2} - \frac{\hbar^2}{mr} \frac{dR}{dr} + \frac{l(l+1)}{2mr^2} R + V(r) R = ER$$

R For bound states $R \rightarrow 0$ as $r \rightarrow \infty$

R, $\frac{dR}{dr}$ should be finite and continuous everywhere.

$$-\frac{\hbar^2}{2m} \frac{d^2 R}{dr^2} - \frac{\hbar^2}{mr} \frac{dR}{dr} + \frac{l(l+1)\hbar^2}{2mr^2} R + VR = ER$$

$$R(r) = \frac{u}{r}$$

Since R is finite everywhere, $u \rightarrow 0$ as $r \rightarrow 0$ faster than r or as r

$$\frac{dR}{dr} = \frac{1}{r} \frac{du}{dr} - \frac{1}{r^2} u$$

$$\frac{d^2 R}{dr^2} = \frac{1}{r} \frac{d^2 u}{dr^2} - \frac{2}{r^2} \frac{du}{dr} + \frac{2}{r^3} u$$

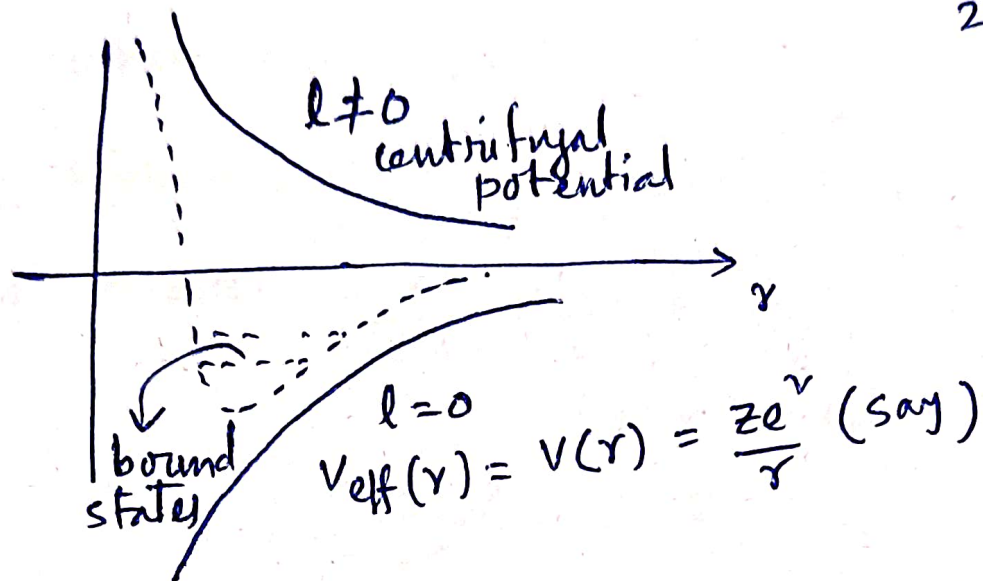
$$-\frac{\hbar^2}{2m} \left[\frac{1}{r} \frac{d^2 u}{dr^2} - \frac{2}{r^2} \frac{du}{dr} + \frac{2}{r^3} u \right] - \frac{\hbar^2}{mr} \left[\frac{1}{r} \frac{du}{dr} - \frac{1}{r^2} u \right] + \frac{l(l+1)\hbar^2}{2mr^3} u + V(r) \frac{u}{r} = E \frac{u}{r}$$

$$\Rightarrow -\frac{\hbar^2}{2m} \frac{1}{r} \frac{d^2 u}{dr^2} + \frac{l(l+1)\hbar^2}{2mr^3} u + V(r) \frac{u}{r} = E \frac{u}{r}$$

$$\Rightarrow -\frac{\hbar^2}{2m} \frac{d^2 u}{dr^2} + \left[\frac{l(l+1)\hbar^2}{2mr^2} + V(r) \right] u = E u$$

$$\left[\text{in classical mechanics } V_{\text{eff}} = V(r) + \frac{l^2}{2mr^2} \right]$$

Effective potential $V_{\text{eff}}(r) = V(r) + \frac{l(l+1)\hbar^2}{2mr^2}$



$$\Psi(r, \theta, \phi) = \frac{u(r)}{r} Y_{lm}(\theta, \phi)$$

(u depends on l)

$$\begin{aligned} \int |\Psi|^2 d^3x &= \int_0^\infty r^2 dr \int d\Omega Y_{lm}^*(\theta, \phi) Y_{lm}(\theta, \phi) \\ &= \int_0^\infty dr |u(r)|^2 = 1 \end{aligned}$$

orthonormal
eigen functions
of L^2, L_z

Ψ is normalized if u is normalized.

$$\begin{aligned} \text{(Also } \int d\Omega Y_{l'm'}^*(\theta, \phi) Y_{lm}(\theta, \phi) \\ = \delta_{l',l} \delta_{m',m} \text{)} \end{aligned}$$

For Coulomb potential, as $r \rightarrow 0$, the centrifugal term dominates. Also since $u = rR$ (R finite at $r \rightarrow 0$)

$$\text{As } r \rightarrow 0, \quad -\frac{\hbar^2}{2m} \frac{d^2 u}{dr^2} + \frac{l(l+1)\hbar^2}{2mr^2} u = 0$$

$$\Rightarrow \frac{d^2 u}{dr^2} = \frac{l(l+1)}{r^2} u \Rightarrow u = r^{l+1} \text{ or } u = r^{-l}$$

as $r \rightarrow 0$ $u = r^{-l}$ is not normalizable
 $u(r) \sim r^{l+1}$ let us verify!

$$-\frac{\hbar^2}{2m} \frac{d^2 u}{dr^2} + V_{\text{eff}}(r) u = E u$$

$$\text{where } V_{\text{eff}} = V(r) + \frac{l(l+1)\hbar^2}{2mr^2}$$

$$V(r) \rightarrow -\frac{1}{r} \text{ (suppose) } \Rightarrow \text{centrifugal term dominates as } r \rightarrow 0$$

$$\text{Eqn becomes } -\frac{\hbar^2}{2m} \frac{d^2 u}{dr^2} + \frac{l(l+1)\hbar^2}{2mr^2} u = 0$$

(Also $u \rightarrow 0$ as $r \rightarrow 0$)

$$\frac{d^2 u}{dr^2} - \frac{l(l+1)}{r^2} u = 0$$

$$\text{Try, } u \sim r^\alpha \Rightarrow \alpha(\alpha-1) - l(l+1) = 0$$

$$\alpha = -l, \alpha = l+1$$

$$u \text{ as } r \rightarrow 0 \quad u(r) \sim r^{-l} \text{ or } r^{l+1}$$

$$\text{Since, } \frac{u(r)}{r} \rightarrow \text{finite as } r \rightarrow 0$$

$$\Rightarrow r^{-l} \text{ soln is not acceptable}$$

$$u(r)(r \rightarrow 0) \sim r^{l+1}$$

$$R(r)(r \rightarrow 0) \sim r^l$$

A constant non-zero wave function at the origin is possible only for $l=0$. For $l \neq 0$, the centrifugal "barrier" prevents the particle from reaching the origin. or in other words $R \rightarrow 0$ when $l \neq 0$.

General soln of Schrodinger eqn for spherically symmetric (central) potential

$$-\frac{\hbar^2}{2m} \frac{d^2 u}{dr^2} + \left[\frac{l(l+1)\hbar^2}{2mr^2} + V(r) \right] u = E u$$

where $R(r) = \frac{u}{r}$, $\psi(\vec{r}) = R(r) Y_{lm}(\theta, \phi)$
 $= \frac{u(r)}{r} Y_{lm}(\theta, \phi)$

$$u \rightarrow r^{l+1} \text{ as } r \rightarrow 0$$

Hydrogenic atom $\rightarrow$ any atom/ion with a single valence electron.

Two body problem involving a positively charged nucleus Ze and a single valence electron of charge $-e$.

The problem can be reformulated into one centre-of-mass degree of freedom that behaves like a free particle and a relative motion degree of freedom which feels the central potential.

$$\frac{\hat{\vec{P}}^2}{2m} \Psi_{CM}(\vec{X}) = E_{CM} \Psi_{CM}(\vec{X})$$

$\hat{\vec{P}} \rightarrow$ total momentum operator $\vec{P} = \vec{P}_p + \vec{P}_e$

$$\left[\frac{\hat{\vec{p}}^2}{2\mu} + V(|\vec{x}|) \right] \Psi_{rel}(\vec{x}) = E_{rel} \Psi_{rel}(\vec{x})$$

where $\mu = \frac{m_e m_p}{m_e + m_p}$ and $\vec{P} = \mu \left(\frac{\vec{P}_e}{m_e} + \frac{\vec{P}_p}{m_p} \right)$

$$\vec{X} = \frac{m_e \vec{X}_e + m_p \vec{X}_p}{m_e + m_p}, \quad \vec{x} = \vec{X}_e - \vec{X}_p$$

$$V(r) = -\frac{Ze^2}{r}$$

$z \rightarrow$ number of protons in the nucleus

For Hydrogen, $z=1$, $z>1$: Hydrogenlike atoms

$$-\frac{\hbar^2}{2m} \frac{d^2 u}{dr^2} + \left[\frac{l(l+1)\hbar^2}{2mr^2} - \frac{Ze^2}{r} \right] u = Eu$$

For bound state solution $E < 0 \Rightarrow E = -|E|$

$$-\frac{\hbar^2}{2m} \frac{d^2 u}{dr^2} + \left[\frac{l(l+1)\hbar^2}{2mr^2} - \frac{Ze^2}{r} \right] u = -|E|u$$

In the limit $r \rightarrow \infty \Rightarrow$ neglect $\frac{1}{r^2}, \frac{1}{r}$ terms

$$-\frac{\hbar^2}{2m} \frac{d^2 u}{dr^2} = -|E|u \Rightarrow \boxed{\frac{d^2 u}{dr^2} - \frac{2m|E|}{\hbar^2} u = 0}$$

$$\text{put } \beta^2 = \frac{2m|E|}{\hbar^2}$$

$$u(r) \approx e^{\pm \beta r}$$

$$R \rightarrow 0 \text{ as } r \rightarrow \infty \Rightarrow R = \frac{u}{r} \rightarrow 0 \Rightarrow u \rightarrow 0 \text{ as } r \rightarrow \infty$$

$$u(r) \approx e^{-\beta r}$$

$$\text{General soln: } u_l(r) = \left[\sum_{i=1}^n a_i r^i \right] e^{-\beta r}$$

Recall u is dependent on l for $l \neq 0$

The polynomial is finite $i=1$ to n , since otherwise the summation term will blow up as $n \rightarrow \infty$

The lowest value of $i=1$ since $u \rightarrow 0$ as $r \rightarrow 0$

$$\underline{\underline{l=0}} \quad -\frac{\hbar^2}{2m} \frac{d^2 u}{dr^2} - \frac{ze^2}{r} u = Eu$$

$$\underline{n=1} \quad u = r e^{-\beta r} \Rightarrow \frac{d^2 u}{dr^2} = -2\beta e^{-\beta r} + \beta^2 r e^{-\beta r}$$

$$\Rightarrow -\frac{\hbar^2}{2m} [-2\beta + \beta^2 r] e^{-\beta r} - \frac{ze^2}{r} r e^{-\beta r} = E r e^{-\beta r}$$

$$\Rightarrow \left(\frac{\hbar^2 \beta}{m} - ze^2 \right) - \frac{\hbar^2 \beta^2}{2m} r = E r$$

Comparing the coefficients of r on either side

$$E = -\frac{\hbar^2 \beta^2}{2m} \Rightarrow \text{consistent with } \beta = \frac{2m|E|}{\hbar^2}$$

$$\text{Also, } \frac{\hbar^2 \beta}{m} - ze^2 = 0 \Rightarrow \frac{\hbar^2}{m} \sqrt{\frac{2m|E|}{\hbar^2}} = ze^2$$

$$\Rightarrow |E| = \frac{m z^2 e^4}{2 \hbar^2}$$

$$\text{Bohr radius } a_0 = \frac{\hbar^2}{m e^2} = 0.529 \text{ \AA}$$

$$\Rightarrow |E| = \frac{e^2}{2a_0} z^2$$

$$\text{Again, } \frac{e^2}{2a_0} = \frac{1}{2} e^2 \left(\frac{m e^2}{\hbar^2} \right) = \frac{1}{2} \frac{e^4}{\hbar^2 c^2} m c^2$$

$$\text{Fine structure constant } \alpha = \frac{e^2}{\hbar c} \approx \frac{1}{137}$$

$$\Rightarrow \frac{e^2}{2a_0} = \frac{1}{2} \alpha^2 m c^2, \quad m c^2 \rightarrow \text{rest mass energy of electron} \approx 0.511 \text{ MeV}$$

$$\Rightarrow \frac{e^2}{2a_0} = \frac{1}{2} \left(\frac{1}{137} \right)^2 \times 511 \times 10^3 \text{ eV} \approx 13.6 \text{ eV}$$

$$\boxed{E = -|E| = -13.6 z^2 \text{ eV}}$$

Ground state energy of hydrogenic atom

$$\text{For Hydrogen } z=1 \Rightarrow E_g = -13.6 \text{ eV}$$

$$\underline{l=0, n=1} \quad U = C_N r e^{-\beta r} \Rightarrow R = \frac{U}{r} = C_N e^{-\beta r}$$

$$\Psi = C_N e^{-\beta r} Y_{00}(\theta, \phi) \quad [l=0, m=0]$$

$$\beta = \sqrt{\frac{2m|E|}{\hbar^2}} = \sqrt{\frac{me^2 z^2}{a_0 \hbar^2}} \quad [\text{since } |E| = \frac{e^2 z^2}{2a_0}]$$

$$\text{For } z=1, \quad \beta = \sqrt{\frac{me^2}{a_0 \hbar^2}}$$

$$\text{Now } a_0 = \frac{\hbar^2}{me^2} \Rightarrow \beta = \frac{1}{a_0}$$

$$\text{Normalize: } 4\pi |C_N|^2 \int_0^\infty r^2 e^{-2r/a_0} dr = 1$$

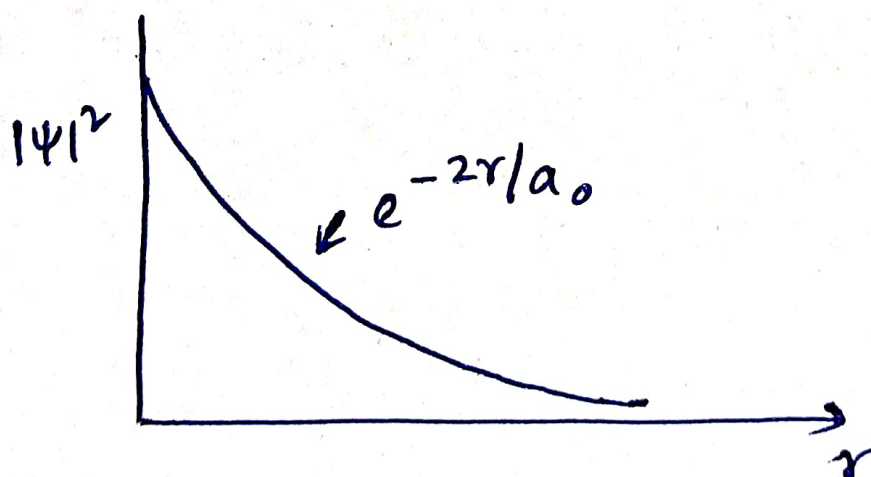
$$\Rightarrow 4\pi |C_N|^2 \frac{a_0^3}{8} \int_0^\infty y^2 e^{-y} dy = 1 \quad [\text{put } y = \frac{2r}{a_0}]$$

$$\Rightarrow 4\pi |C_N|^2 \frac{a_0^3}{8} \cdot 2! = 1$$

$$\Rightarrow C_N = \frac{1}{\sqrt{\pi a_0^3}}$$

Hence the normalized ground state wave function for hydrogen is

$$\Psi = \frac{1}{\sqrt{\pi a_0^3}} e^{-r/a_0}$$



A more rigorous approach:

The radial Schrodinger eqn for bound states $E < 0$

$$\left(-\frac{\hbar^2}{2m} \frac{d^2}{dr^2} + \frac{l(l+1)\hbar^2}{2mr^2} - \frac{Ze^2}{r} \right) u = Eu$$

We define a unit-free, dimensionless co-ordinate

$$\gamma = \frac{a_0}{2Z} x$$

$$\Rightarrow \left(-\frac{\hbar^2}{2m} \frac{4Z^2}{a_0^2} \frac{d^2}{dx^2} + \frac{4Z^2}{a_0^2} \frac{\hbar^2}{2m} \frac{l(l+1)}{x^2} - \frac{2Z^2 e^2}{a_0} \frac{1}{x} \right) u = Eu$$

$$\Rightarrow \left[\frac{2\hbar^2 Z^2}{ma_0^2} \left(-\frac{d^2}{dx^2} + \frac{l(l+1)}{x^2} \right) - \frac{2Z^2 e^2}{a_0} \frac{1}{x} \right] u = Eu$$

$$\Rightarrow \text{Here } \frac{2\hbar^2 Z^2}{ma_0^2} = \frac{2\hbar^2 Z^2}{ma_0} \frac{me^2}{\hbar^2} = \frac{2Z^2 e^2}{a_0}$$

$$\Rightarrow \left[-\frac{d^2}{dx^2} + \frac{l(l+1)}{x^2} - \frac{1}{x} \right] u = \frac{E}{\left(\frac{2Z^2 e^2}{a_0} \right)} u$$

Define a dimensionless parameter $k^2 = \frac{-E}{\left(\frac{2Z^2 e^2}{a_0} \right)} > 0$

$$\Rightarrow \left[-\frac{d^2}{dx^2} + \frac{l(l+1)}{x^2} - \frac{1}{x} \right] u = -k^2 u$$

We further define a dimensionless co-ordinate ρ

$$\rho = kx = \frac{2kZ}{a_0} r$$

$$\Rightarrow \left[-\frac{d^2}{d\rho^2} + \frac{l(l+1)}{\rho^2} - \frac{1}{k\rho} \right] u = -u$$

As $\rho \rightarrow 0$, $u \rightarrow 0$ and $\frac{l(l+1)}{\rho^2}$ dominates over $\frac{1}{k\rho}$

$$\Rightarrow \frac{d^2 u}{d\rho^2} - \frac{l(l+1)}{\rho^2} u = 0 \Rightarrow u \sim \rho^{l+1}$$

As $\rho \rightarrow \infty$, $u = e^{-\rho}$ as before.

General soln $u(\rho) = \rho^{l+1} w(\rho) e^{-\rho}$

Plugging the expression back in the eqn, one gets a differential eqn for w .

$$\rho \frac{d^2 w}{d\rho^2} + 2(l+1-\rho) \frac{dw}{d\rho} + \left[\frac{1}{k} - 2(l+1) \right] w = 0$$

We express w as a series expansion $w = \sum_{k=0}^{\infty} a_k \rho^k$

$$\frac{dW}{dp} = \sum_{k=0}^{\infty} a_k k p^{k-1}; \quad \frac{d^2 W}{dp^2} = \sum_{k=0}^{\infty} a_k k(k-1) p^{k-2}$$

$$\Rightarrow p \sum_{k=0}^{\infty} a_k k(k-1) p^{k-2} + 2[(l+1)-p] \sum_{k=0}^{\infty} a_k k p^{k-1} + \left[\frac{1}{k} - 2(l+1)\right] \sum_{k=0}^{\infty} a_k p^k = 0$$

collect the coefficients of p^k and equate to zero.

$$a_{k+1}(k+1)k + 2(l+1)a_{k+1}(k+1) - 2a_k k + \left[\frac{1}{k} - 2(l+1)\right]a_k = 0$$

$$\Rightarrow a_{k+1}[(k+1)k + 2(l+1)(k+1)] = [2k - \frac{1}{k} + 2(l+1)]a_k$$

$$\Rightarrow \frac{a_{k+1}}{a_k} = \frac{[2(k+l+1) - \frac{1}{k}]}{[(k+1)(k+2l+2)]}$$

The series must ~~converge~~ terminate for wave function to be normalizable. In the large k limit, we can write,

$$\frac{a_{k+1}}{a_k} \approx \frac{2}{k+1} \Rightarrow a_{k+1} = \frac{2}{k+1} a_k$$

$$k=0: \quad a_1 = 2a_0 = \frac{2^1}{1!} a_0$$

$$k=1: \quad a_2 = a_1 = 2a_0 = \frac{2^2}{2!} a_0$$

$$k=2: \quad a_3 = \frac{2}{3} a_2 = \frac{4}{3} a_0 = \frac{2^3}{3!} a_0$$

$$k=3: \quad a_4 = \frac{2}{4} a_3 = \frac{2}{4} \cdot \frac{4}{3} a_0 = \frac{2^4}{4!} a_0$$

$$k=4: \quad a_5 = \frac{2}{5} a_4 = \frac{2^5}{5!} a_0$$

$$\Rightarrow \boxed{a_k = \frac{2^k}{k!} a_0}$$

$$W = \sum_{k=0}^{\infty} a_k \rho^k = a_0 \sum \frac{2^k}{k!} \rho^k = a_0 e^{2\rho}$$

(b) $\psi(\rho)$ is unnormalizable. The series has to be truncated

Suppose W is a polynomial of degree ~~N~~ N so the coefficients satisfy

$$a_N \neq 0 \text{ and } a_{N+1} = 0$$

$$\text{This implies } \frac{1}{k} = 2(N+l+1)$$

k encodes information about E

E is quantized $\Rightarrow$ related to integers

$$l = 0, 1, 2, \dots$$

$$N = 0, 1, 2, \dots$$

For each combination of ~~ener~~ N, l we have an expression for energies

We define the principal quantum number

$$n = N + l + 1 = \frac{1}{2k}, \quad l = 0, 1, 2, \dots$$

$$N = 0, 1, 2, \dots$$

$$n \geq 1$$

$$k = \frac{1}{2n}$$

$$k^2 = \frac{-E}{\left(\frac{2Z^v e^v}{a_0}\right)} \Rightarrow E = -\frac{2Z^v e^v}{a_0} k^2$$

$$\text{Again } k = \frac{1}{2n} \Rightarrow E = -\frac{2Z^v e^v}{4a_0} \frac{1}{n^2}$$

$$\boxed{E_n = -\frac{Z^v e^v}{2a_0} \frac{1}{n^2}}$$